# A Short Theoretical Note on Identity-Tanh-Offdiag Residual Mixing

**Abstract**

This note gives a simple theoretical interpretation of the `identity_tanh_offdiag` parameterization for multi-stream hyper-connections. The main claim is that this parameterization preserves the identity residual path exactly, while allowing bounded signed communication between streams. This can improve optimization stability relative to fully dynamic mHC and can scale more naturally to larger numbers of streams than Sinkhorn-based mHC or permutation-mixture mHC-lite.

## 1 Setup

Let $n$ be the number of residual streams. Standard mHC learns a residual mixing matrix $H_{\text{res}} \in \mathbb{R}^{n\times n}$ and constrains it to be doubly stochastic, usually by applying Sinkhorn-Knopp normalization to learned logits. mHC-lite instead represents $H_{\text{res}}$ as a convex combination of permutation matrices:

$$H_{\text{res}} = \sum_{\pi\in S_n} a_\pi P_\pi, \qquad a_\pi \geq 0, \qquad \sum_{\pi\in S_n} a_\pi = 1. \tag{1}$$

Both methods keep $H_{\text{res}}$ inside the Birkhoff polytope.

The proposed `identity_tanh_offdiag` parameterization is

$$H_{\text{res}} = I + \gamma(1 - I) \odot \tanh(W), \tag{2}$$

where $W \in \mathbb{R}^{n\times n}$ is trainable, $\gamma > 0$ is a small scale parameter, and $(1-I)$ masks out the diagonal. Thus

$$(H_{\text{res}})_{ii} = 1, \qquad (H_{\text{res}})_{ij} \in [-\gamma, \gamma] \quad \text{for } i \neq j. \tag{3}$$

## 2 Main Difference

The key distinction is geometric. mHC and mHC-lite learn a matrix in the Birkhoff polytope. In that geometry, the identity matrix is a vertex. Moving away from identity requires transferring mass from the diagonal to off-diagonal entries, because each row and column must still sum to one. Therefore stronger cross-stream mixing necessarily weakens the self-stream identity path.

In contrast, Equation (2) keeps the diagonal fixed at one. Off-diagonal terms are added as small signed perturbations around the identity. The residual stream keeps its direct identity route, while cross-stream communication is learned as a bounded correction.

## 3 Local Identity Stability

**Proposition 1.** *Suppose each layer uses a residual mixing matrix $H_\ell = I + E_\ell$ with $\|E_\ell\|_2 \le \epsilon_\ell$. Then, for small $\epsilon_\ell$,*

$$\left\| \prod_{\ell=1}^{L} H_\ell \right\|_2 \le \exp\left( \sum_{\ell=1}^{L} \epsilon_\ell \right). \tag{4}$$

This follows from submultiplicativity:

$$\left\| \prod_{\ell=1}^{L} (I + E_\ell) \right\|_2 \le \prod_{\ell=1}^{L} (1 + \epsilon_\ell) \le \exp\left( \sum_{\ell=1}^{L} \epsilon_\ell \right). \tag{5}$$

For `identity_tanh_offdiag`, $\|E_\ell\|_2$ is controlled by $\gamma$ and the off-diagonal pattern. Thus a small $\gamma$ gives a direct mechanism for preserving stable residual signal propagation across depth.

## 4 Optimization Advantages

Near initialization, if $W_{ij} = 0$ for $i \neq j$, then

$$\frac{\partial (H_{\text{res}})_{ij}}{\partial W_{ij}} = \gamma. \tag{6}$$

Thus every off-diagonal direction starts with a nonzero and well-controlled gradient. This is useful when the model only needs weak cross-stream communication.

By comparison, in mHC and mHC-lite the off-diagonal directions are mediated by normalization maps such as Sinkhorn-Knopp or softmax over permutation coefficients. When initialized close to identity, off-diagonal probabilities can be very small. Learning a useful off-diagonal component may then require moving through a saturated probability simplex region, where gradients are attenuated and highly coupled across entries.

Another important difference is sign. Birkhoff-constrained matrices are nonnegative, so they implement convex averaging between streams. The `identity_tanh_offdiag` matrix permits signed off-diagonal entries. This allows one stream to subtract features from another, not only average them. This signed correction can be more expressive while still remaining bounded.

## 5 Why Signed Off-Diagonal Entries

We do not restrict off-diagonal entries to be nonnegative. The motivation is the expressivity argument made by Spectral-Sphere-Constrained Hyper-Connections (sHC) [3]. SHC identifies the Birkhoff-polytope constraint used by mHC and mHC-lite as having an expressivity bottleneck: because entries are nonnegative, residual streams are restricted to convex mixing. This prevents subtractive feature disentanglement and limits active cross-stream diversification.

In the notation of our parameterization,

$$y_i = x_i + \sum_{j \neq i} \delta_{ij} x_j, \qquad \delta_{ij} \in [-\gamma, \gamma]. \tag{7}$$

Allowing $\delta_{ij} < 0$ gives the residual mixing matrix the same kind of signed interaction space emphasized by sHC, while the exact identity diagonal and the bound $|\delta_{ij}| \le \gamma$ keep the method local around the ordinary residual path:

$$\delta_{ij} = \gamma \tanh(W_{ij}), \qquad i \neq j. \tag{8}$$

Thus, the decision to allow signed off-diagonal entries is not an unconstrained relaxation of mHC. It is a local identity-centered way to inherit the specific expressivity benefit highlighted by sHC: negative residual-mixing entries enable subtractive interactions that nonnegative convex mixing cannot represent.

## 6 Scalability With More Streams

The scalability difference is mainly parameteric and algorithmic.

For dynamic mHC, generating a full dynamic $H_{\text{res}}$ requires producing $n^2$ residual-mixing logits from a representation whose width also scales with $n$. This creates a large cost as $n$ grows. Sinkhorn-Knopp also adds iterative normalization cost.

For mHC-lite, the exact permutation-mixture representation requires all $n!$ permutation matrices. This is practical for $n = 4$, where $n! = 24$, but quickly becomes infeasible for $n = 16$ or $n = 32$.

By contrast, `identity_tanh_offdiag` only stores a static $n \times n$ matrix per hyper-connection module, and it does not need Sinkhorn iterations or permutation enumeration. If dynamic residual mixing is disabled, the dynamic network only predicts the pre-branch routing weights. This gives a much milder scaling path for large stream counts.

## 7 Why mHC and mHC-lite Often Converge Toward Identity

Empirically, mHC and mHC-lite residual matrices often become close to identity: diagonal entries approach one and off-diagonal entries remain small. This behavior is consistent with the theory above.

First, the identity path is the safest solution for deep residual networks. It preserves information and gradient flow across depth. Second, in the Birkhoff polytope, increasing off-diagonal mass necessarily decreases diagonal mass. Unless cross-stream averaging gives a large loss reduction, optimization favors staying near identity. Third, attention and MLP parameters can already perform most feature transformation, so $H_{\text{res}}$ mainly needs to stabilize the residual transport rather than learn an aggressive mixing rule.

This suggests that the useful role of $H_{\text{res}}$ is often not to learn a free global transport matrix, but to provide stable identity transport plus small corrective communication. The `identity_tanh_offdiag` parameterization encodes exactly this inductive bias.

## 8 Interpretation

The proposed method can be summarized as replacing a global transport constraint with a local perturbation model:

$$\text{Birkhoff transport} \quad \longrightarrow \quad \text{identity transport} + \text{bounded signed off-diagonal correction.} \tag{9}$$

This explains why it may train better at the same stream count and why it can scale to larger stream counts. It preserves the main residual highway, provides non-degenerate gradients for cross-stream communication, permits signed corrections, and avoids the computational bottlenecks of Sinkhorn iterations or permutation enumeration.

## References


[1] DeepSeek-AI. *mHC: Manifold-Constrained Hyper-Connections*. arXiv:2512.24880.

[2] *mHC-lite: You Don't Need 20 Sinkhorn-Knopp Iterations*. arXiv:2601.05732.

[3] Zhaoyi Liu, Haichuan Zhang, and Ang Li. *Beyond the Birkhoff Polytope: Spectral-Sphere-Constrained Hyper-Connections*. arXiv:2603.20896.